



“Electronic signals can be used to alter the workings of the human brain in a variety of ways”

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Human-Machine Convergence

If asked to name the two technical advances that will have a profound effect on society in the not-too-distant future, my answer would be clear and certain.

Firstly, the ability to communicate directly from brain to brain between humans, which could be called ‘Thought Communication’. Secondly, the use of electronic signals, as opposed to chemical, for the treatment of all sorts of ailments, particularly those that are mental in nature, which could be described as ‘E-Medicine’.

These two seemingly disparate realisations have, perhaps surprisingly, a common technological base—namely the use of implant technology to link the human brain directly into the computer network. Essentially, humans will become nodes on the Internet. From a cybernetic viewpoint, signals on the human nervous system and those in the computer network are interchangeable, and are effectively one and the same.

The two major advances of Thought Communication and E-medicine also have commonality, in the sense, important initial steps have been taken in the appropriate direction. In both cases, we are already on the way to achieving the results.

In the case of Thought Communication, contact has been successfully achieved thus far, in the form of radio telegraphic communication that takes place directly between the nervous systems of two humans.

An experiment repeated between two human brains is merely a matter of time. What will be interesting then is how swiftly things will move forward. Will it be a case of telephonic communication, followed by memory uploads and downloads, or will we all learn to think in a new language—with our original spoken language, with speech (in general) becoming obsolete?

It's important to realise how poorly humans communicate at present, converting complex electrochemical brain signals into relatively trivial bursts of pressure waves, by mouth movement, which is called speech. These waves are picked up in the ears of other individuals, who in turn, try to put some sense into the signals they are receiving.

Unfortunately, the understood message is invariably altered from the original, intended message, especially when different cultures are involved. Thought Communication, therefore, offers a much more powerful, effective, and less error-prone means of communication.

As was the case with the telegraph, the telephone and the mobile phone—all had a significant impact on society, culture and commercial interests. Thus, communicating by thought is likely to have an even more profound effect than all the above three combined.

E-Medicine also depends on the basic electrochemical nature of the human brain and nervous system. Chemicals have, for some reason, been the mainstay in the Western world for treatment of all types of diseases for centuries. However, in the twilight of the last millennium we witnessed the first twinklings of the successful treatment of diseases by means of electronic brain stimulation through implanted electrodes.

The treatment of Parkinson's disease by this approach was found to be very successful. In fact, in the new millennium, the number of patients so treated had already run into hundreds. Now we can witness reports on successful trials and further studies involving individuals with epilepsy or multiple sclerosis. Clearly, the field is rapidly growing and diversifying.

Indeed, it is exciting to look at the range of absolute potential in the broader sense of E-Medicine. Basically, what we are looking at is the use of